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Implementing a Points-Based Valuation System for Property Taxation

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Nicolas Orgeira Pillai, Xaver Schenker, Wilson Prichard and Graeme Stewart-Wilson

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Acronyms

> AVM	Automated Valuation Model
> GIS	Geographic Information System
> HFC	High Frequency Check
> OLS	Ordinary Least Squares
> UN-OCHA	United Nations Office for the Coordination of Humanitarian Affairs

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1 Introduction

Property taxation is typically defined as a recurrent tax on the value of a property. Taxing properties based on value ensures that the overall system remains equitable, as more valuable properties—presumably owned by wealthier residents—receive higher tax bills than less valuable properties. Taxing properties based on value also helps ensure the tax base is naturally buoyant, as revenue collection will increase with rising property values. In practice, however, determining the value of properties can be a highly complex exercise. The true value of a property can only be known when it is sold on the open market and is thus not directly observable for most properties at the time of taxation. For this reason, determining the taxable value of properties inevitably requires an administrative judgement about the most appropriate value to use for taxation.

Different approaches to making this administrative judgement can be roughly categorized along a spectrum according to their simplicity and equitability. At one end of the spectrum, a generic flat rate per property—effectively independent of market value—is the simplest approach. But it is also highly regressive, as low-value properties will be charged disproportionately more than high-value properties. Area-based approaches provide a slightly more equitable option than the flat rate approach. The value of a property—and the tax to be paid—are calculated based on property size, perhaps adjusted depending on its location. Hence, the larger a property, the higher the tax owed to the government, and vice versa. Area-based approaches require collecting data on property sizes and are thus more administratively demanding than a flat rate. Ultimately, however, both flat rate and area-based approaches result again in lower-value properties paying disproportionately higher taxes in comparison to higher-value properties of the same size.

On the other end of the spectrum, taxing properties based on their market values is—in theory—the most equitable approach. However, determining market values is a demanding task, especially in underdeveloped or opaque real estate markets. If properties are rarely bought and sold—or if sale prices are not publicly or reliably available when transactions do occur—then little basis exists on which to estimate the market value of properties. In such contexts market valuation can become subjective and susceptible to collusion or rent-seeking practices.

Increasing digitalization of property taxation is creating new opportunities to systematize valuation to maximize equitability while minimizing the administrative challenges of market-based assessment. Automated valuation models (AVMs), for instance, are a class of tools that estimate property values using a formula. Inputs to the formula are objective property characteristics (e.g., location, built area, type of walls, type of roof, year of construction, floor area, etc.) along with market data like sale prices for similar properties, construction costs, and projected income generated by the property. The underlying formula is calibrated with statistical methods (regression analysis¹), and its output is an estimated market value that can be used for taxation.

The primary benefit of AVMs is that value estimates can be generated for many properties at once by simply inputting data on the characteristics of those properties. That reduces the scope for errors, collusion, and rent-seeking, and also ensures consistency.

¹ Regression analyses determine the relationship between the mean value of an output variable, and the corresponding values of one or several input variables.

AVMs cover a wide range of potential approaches. These approaches themselves can be roughly organized along a spectrum according to the number and sophistication of the variables included in their formula, and the complexity of the formula itself. As the number and sophistication of variables increase, the overall cost and complexity of implementing the model also increase. Higher complexity models may deliver more accurate results, but also increase the technical expertise required to operate the system, and likely decrease transparency and comprehensibility to taxpayers.

This guidance note focuses on the implementation of AVMs in contexts where property market data is hard to access, inconsistent, or unrepresentative, where administrative capacity is relatively limited, and where risks of collusion are high. The approach presented emphasizes data-gathering procedures that are simple and verifiable, and it prioritizes formulas that are also relatively simple and transparent. This simplified valuation method—often referred to as “points-based valuation”—combines the strengths of area-based and market value-based assessment to implement a valuation system that is as equitable and transparent as possible while remaining relatively simple to administer and maintain.

Like all AVMs, the simplified valuation approach uses a formula to calculate a proxy for the value of properties. Inputs to the formula include property size, location, neighbourhood features, and externally observable building characteristics. The formula takes property size as the initial input, and then makes value adjustments based on where the property is located, the features of its surrounding environment, and the property’s externally visible characteristics. The goal of the simplified valuation approach is not to arrive at perfectly accurate market values, but rather to generate good and consistent approximations of value. The emphasis is on ensuring that the rank order of property values generated by the simplified method approximate their relative values in the market, thus ensuring fairness and equity.

The terminology of “points-based valuation” is a reminder that actual “market value” is inherently unknown and imprecise, particularly in countries with poorly developed and opaque property markets. Instead, the method can be understood as assigning additional “points” to properties with characteristics that, on average, increase property value. Ideally, the “points” assigned to properties are calibrated with actual market value data to maximize accuracy, make estimated values easier to understand for taxpayers, and ensure revenue buoyancy over time.

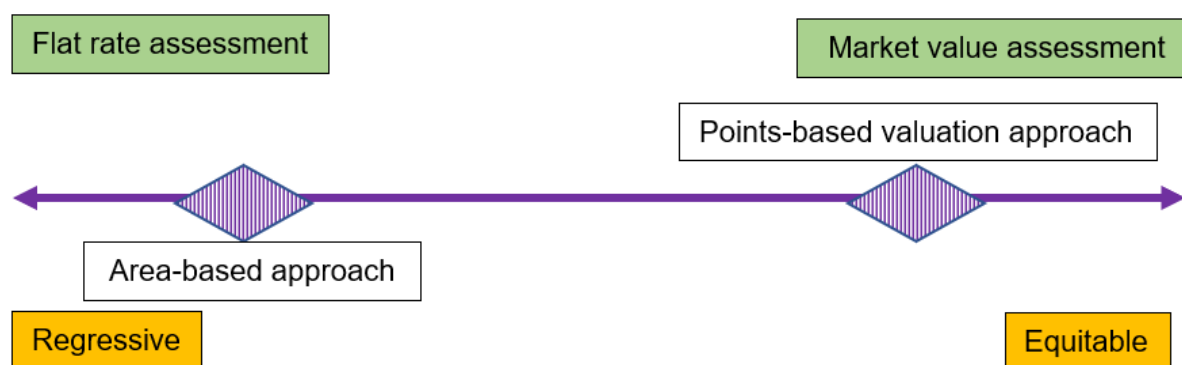


Figure 1 Equitability of property tax approaches

This guidance note describes the steps required to implement a points-based valuation system, including the various considerations and trade-offs that can help adapt the approach to different contexts. The next section discusses the criteria included in the valuation model. For each component, the guidance note describes the logic of what is needed, any key considerations or trade-offs, and how it could be implemented in practice. Section three discusses the main decisions that need to be made to organize data collection so that every property in the jurisdiction is included in the tax net. Section four outlines steps to gather the market information needed to calibrate the valuation model, including trade-offs and different options in approach. Sections five and six describe how to build a valuation model with the data collected, and how to update and adjust that model to reflect the needs of specific contexts. The final section outlines how to apply the model to property data to calculate estimated values that can be used for taxation.

2 Defining Components of the Valuation Model

The simplified valuation approach explained in this guidance note is applied easily to residential properties, which account for most properties in most jurisdictions. Commercial properties can present challenges to the simplicity of the model because their construction may be idiosyncratic, making them harder to compare to each other and determine relative property values. For example, neither a bank with its increased security features and a vault, nor a warehouse with very high ceilings and large open spaces, are directly comparable to the characteristics of a typical residential property. In general, standard commercial buildings such as shops, workshops, or small warehouses are easily incorporated into the simplified valuation model because their style of construction is roughly comparable to that of residential properties. More distinct commercial properties, however, may require a distinct valuation approach. Three broad strategies to assessing commercial properties are discussed in Appendix 1. The remainder of this guidance note focuses on implementing a simplified valuation approach for residential properties.

Property taxation first requires that all objects subject to the tax are accurately identified and located. The basis for any property tax system is therefore a complete and up-to-date geospatial register of all properties in the jurisdiction. This guidance note assumes that—prior to implementing the points-based valuation system—a clear plan for updating or creating a new geospatial register is already in place. A strategy for creating a comprehensive geospatial register that is compatible with the points-based valuation system is described in a separate guidance note, which should be read before this one.² At a minimum, the geospatial register needs to identify each taxable property in the jurisdiction by GPS coordinates. If the geospatial register is built from aerial imagery, then it may also include measurements for the size of each property in the form of traced building rooftops.

Once a plan for the geospatial register is in place, the next consideration is how to define the key components of the valuation model. The goal of the simplified valuation approach is to take the characteristics of each property and its surrounding environment as inputs, and to output a proxy for property value. As such, deciding on and defining the inputs that shape property value is the first step in creating an effective model.

² *Guidance Note: Property Identification and Mapping for Property Taxation.*

The model criteria should be accurate and simple, and clearly defining them is therefore an iterative process involving discussion and feedback from multiple stakeholders. Forming a dedicated committee of specialists to coordinate these discussions and make final decisions can help ensure that sufficient time and expertise are devoted to the task. The committee will ideally include representatives from all relevant departments and ministries such as valuation, finance, housing and urban planning, and local government. In-depth knowledge of property markets and construction types and requirements are helpful to ensure a high level of quality in the assessment model.

The main components of the simplified valuation model include property size, location, neighbourhood features, and property characteristics. This section of the guidance note aims to describe at a broad level what is needed for each component, any important trade-offs or considerations, and how each component can be operationalized in practice. It is the task of the specialized committee to decide on the specific definitions for each component that best fit the context of the implementing jurisdiction.

2.1 Eligibility

As a first step, definitions of assessable and non-assessable building structures need to be clarified. Property valuation only applies to assessable building structures and excludes non-assessable structures as they are not eligible for taxation. In many places, definitions of assessable and non-assessable structures are articulated in the relevant legal framework. For instance, many countries exempt some types of buildings from taxation (e.g., those used for religious purposes, public schools, hospitals or foreign embassies). Similarly, structures of a non-permanent nature (e.g., no permanent roof/walls/foundation) or illegal constructions (e.g., in zones where building is not permitted) may also be exempt from property taxation.³

2.2 Property Size

Property size is a core variable in shaping property value. Implementers need to define a clear and consistent methodology for measuring property sizes that applies across all property types. In some countries, property size is inclusive of both buildings and land holdings, while in others property taxation applies only to the size of built structures. What follows focuses on built structures, with the possibility of applying equivalent methods of mapping and measuring land.

As described in the Property Identification and Mapping Guidance Note,⁴ given the current state of technology the most cost- and time-effective approach to measuring properties is likely tracing building rooftops from aerial imagery. This approach allows technicians to measure many properties at once in a desk-based exercise, as opposed to time-consuming manual measurement of building footprints, while at the same time georeferencing properties for ease of navigation and identification. The number of floors of each property can then be captured by enumerators during the property survey, with total property size calculated by multiplying rooftop area by the number of floors.

³ *Illegal structures are often exempt from property taxation as taxing such properties may be perceived as legitimizing them.*

⁴ *Guidance Note: Property Identification and Mapping for Property Taxation.*

While roofline measurements from aerial imagery are not exactly equivalent to manual building footprint measurements, the key point for an effective valuation model is that the same approach is applied consistently to all properties. While roofline measurements may be, on average, slightly larger than manual building footprint measurements (because of roof overhangs), provided that all properties in a jurisdiction are measured the same way then the relative distribution of property sizes will remain unchanged between the two methods.⁵

2.3 Geographical Variables

Geographical variables matter because two identical properties in two different areas—for example, one in the central business district and another in the suburbs—will have different values due to where they are located. To account for this difference, the simplified valuation model starts from property size, and then makes value adjustments based on the property's location and the features of its surrounding environment.

Depending on the quality and availability of existing maps, location variables can be coded based on the GPS coordinates of properties. Ideally, the long-term goal of all jurisdictions should be to compile a comprehensive and up-to-date Geographic Information System (GIS) database with clearly delineated value zones, industrial areas, environmental hazards, commercial corridors, school areas, service availability, and other geographical distinctions that capture the drivers of property value. Such a database will enable the coding of geographical variables based on the GPS coordinates of properties. However, where quality data on certain geographical variables are not available, manually collecting information for these variables may be appropriate.

2.3.1 _ Location

Location captures the extent to which, all else equal, some areas are more or less valuable than others. Practically, location is defined by subdividing the jurisdiction into distinct geographic areas using GIS software and allocating a single location to each property based on their GPS coordinates.

These locations should, in theory, be roughly homogenous in terms of property value. That is, similar properties within each location should have roughly similar market values. The question for implementers is how precisely to define and delineate the locations to be used in the valuation model. Two broad approaches can be considered: locations can either be based on pre-existing maps or can be drawn as a separate exercise based on local knowledge of the property market.

⁵ For example, imagine two properties that are identical except for their size: property (a) and property (b). If property (a) has a 20% larger building footprint than property (b), and also has a 20% larger roof area than property (b), then there is no relative difference between relying on manual measurements of building footprints versus remote rooftop measurements from aerial imagery. That is, the relative distribution of value estimates for the two properties will be consistent, regardless of which measurement approach is used. The challenge occurs in cases where property (a) has a larger roof area than property (b), but a smaller building footprint—for example, if property (a) includes a very large, covered veranda but property (b) does not. In this case, the relative distribution of values between the two measurement methods would be inconsistent. However, such cases are likely to be quite rare, and can be addressed during the appeals process. As the automated approach to valuation allows for a detailed list of property features—including measurements—to be easily communicated to taxpayers on their bills, they can verify these measurements and report to the tax authority to adjudicate any inconsistencies.

The simplest approach is to use pre-existing maps, such as political (ward or constituent) boundaries to subdivide the jurisdiction. The benefit of this approach is that such maps are likely readily available and easily adapted to the valuation model. The downside is that political boundaries may not conform well to the distribution of property market values. That is, they may include areas in which the values of similar properties vary widely.

Alternatively, implementers can create more granular neighbourhood or zonal value maps to define locations. These maps can be created by data technicians or urban planners familiar with the local context tracing neighbourhood boundaries in the GIS software. If granular neighbourhood boundaries are difficult to determine from aerial imagery, the exercise can be complemented by walking the neighbourhood boundaries with GPS tracking devices.

Creating neighbourhood value maps is likely more accurate in terms of the distribution of property values, and more intuitive for residents. However, the process also requires deep contextual knowledge of the local property market and urban environment, as well as financial resources, time, and technical capacity to complete the mapping exercise efficiently.

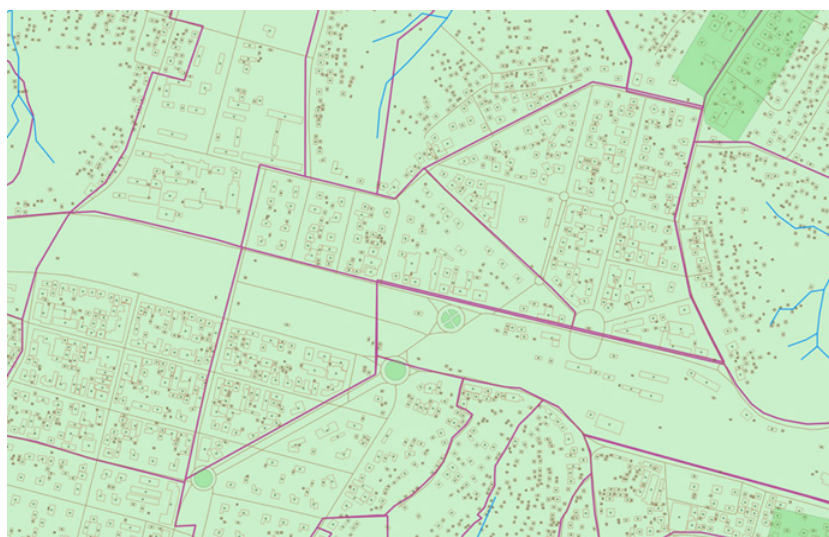


Figure 2 Map of neighbourhood delineations in Kananga, DRC, including rooftop polygons and centroids

→ Special Zones

In addition to location, some jurisdictions may wish to include variables for “special zones” in the valuation model. For instance, properties situated along major commercial corridors, in the central business district, or in important tourist zones may experience a value premium compared to similar properties in nearby areas. Similarly, properties in unplanned settlements or close to environmental hazards⁶ might be discounted compared to similar properties in other areas.

Well-drawn neighbourhood value maps may already capture the impact of these special zones on property value, making the inclusion of special zones in the model redundant. However,

⁶ Adjusting property values based on proximity to environmental hazards can be a delicate matter. If an environmental hazard exists because of government or industrial activity, then there is a legitimate argument to reduce the value of properties in proximity because the hazard is largely outside of residents’ control. However, if the environmental hazard represents a circumscribed area where the government wants to discourage settlement (such as areas prone to flooding or landslides) then they may wish to refrain from allocating value reductions to avoid creating incentives for additional construction in these areas.

if simpler maps are used to define location—like political ward or constituent maps—then including special zones in the valuation model to capture more granular geographical drivers of value can be considered.

The main challenge for implementers is how to define these special zones, and how to collect the data needed to include them in the model. The strategic choice is between either pre-defining these zones in the GIS software or collecting them as part of the property survey. For instance, many jurisdictions will have pre-drawn maps that identify the central business district, unplanned settlements or environmental hazards. If these maps are accurate and accessible to the implementing team, they can be merged with the geospatial property register and automatically assigned to properties based on their GPS coordinates. Even if pre-drawn maps are not available, the implementing team can hire data technicians or urban planners to draw special zone maps in consultation with local authorities knowledgeable about the jurisdiction.

Alternatively, these variables can be collected manually as part of the property survey by asking enumerators to indicate if the property they are surveying is in one of the special zones. This approach will be most appropriate where existing maps are incomplete or out of date, or in cases of insufficient local knowledge to accurately create new maps of special zones.

Manually assigning properties to special zones during data collection has the benefit of offering more fine-grained distinctions but is also likely to require significantly more data cleaning and validation than simply defining special zones in the GIS software and automatically allocating them to properties based on GPS coordinates.

Importantly, the two strategies are not mutually exclusive, as a hybrid approach can be used whereby special zone maps created in GIS are subsequently validated through data collection in the field.

2.3.2 _ Neighbourhood Features

The second group of geographical variables to include in the model are those that capture more granular features of the neighbourhood surrounding each property. For instance, service availability and quality might differ within locations and have implications for the comparative value of properties. The presence of high-quality roads, street lighting, and drainage are examples of neighbourhood features likely to affect property values within locations.

In theory, some jurisdictions may have pre-existing maps of such services that can be automatically merged with property GPS coordinates in the geospatial register. In most cases, however, the presence of these neighbourhood features will need to be collected by enumerators as part of the property survey.

The specialized committee is responsible for deciding on the specific neighbourhood features that are most important in different contexts. Clear definitions also need to be articulated for each neighbourhood feature. For example, enumerators can be instructed to indicate that a property has drainage only if it is directly contiguous with the property's frontage (and not if it is present further down the street). Similarly, clear and precise definitions need to be articulated for other neighbourhood features such as street quality or lighting. As discussed in Section 2.4.3, these definitions should be subjected to rigorous iteration and field testing to ensure that consensus is reached.

2.4 Property Characteristics

While size and geographical variables are generally strong drivers of property value, the simplified model makes subsequent adjustments based on the characteristics of individual properties. Property characteristics are important because even if two properties are the same size and in the same location, their values will differ based on the quality of the property.

Implementers thus need to decide which property characteristics are the main drivers of value and should be included in the model. In theory, the model could include variables for both external and internal building characteristics (such as number of rooms, size of rooms, number of bathrooms, kitchen furnishings, etc.). However, collecting data on internal building characteristics is significantly more costly, slower to implement, harder to verify, and more vulnerable to collusion than an approach that relies exclusively on easily verifiable external building characteristics.

Existing evidence suggests that incorporating internal characteristics has a comparatively minor impact on the overall accuracy of such models. As such, the simplified valuation approach focuses only on external property characteristics that are much easier to collect, update, and verify, which helps to reduce costs, timelines, and risks of collusion or rent-seeking.

2.4.1 _ Defining the List of Property Characteristics

Defining the list of property characteristics will depend on the local knowledge and judgement of the specialized committee but should be informed by several iterative rounds of field testing, as described in Section 2.4.3. The result of this iterative process should be a comprehensive list of building characteristics that enumerators will collect during the property survey. Characteristics included on the list should be capable of roughly describing any property that enumerators will encounter in the field.

To define the property characteristics that will be included in the model, implementers can follow either of two strategies: (1) starting from a list of generic property characteristics,⁷ or (2) starting from a representative sample of property photos.

Starting from a generic list of property characteristics allows implementers to iterate over time, testing the list in the field, and gradually adding or removing characteristics until the final list is roughly consistent with the property characteristics found in a local jurisdiction.

Alternatively, property characteristics can be identified through the analysis of a representative sample of property photos. Every photographed property is described by its characteristics, and the combination of all the descriptions results in an aggregate list of property characteristics divided into various categories.⁸ This aggregate list resembles the generic list used in the former approach but has the advantage of being generated directly from local circumstances.

For both approaches, there is a trade-off between precision and administrability. With more characteristics on the list, enumerators will be able to describe buildings more precisely—but the survey and subsequent analysis will also become more complex and time-consuming. On the other hand, some infrequently appearing characteristics might be important drivers of value (e.g., the presence of a swimming pool) and should be included in the model even though they don't apply to most properties.

⁷ See Appendix 2 for an example generic list of property characteristics.

⁸ A template for this approach can also be found in Appendix 3 of this guidance note.

In general, implementers should aim to strike a balance between precision and simplicity (for instance, by collecting data on the most common types of windows or the dominant wall material) while also ensuring that features that are major determinants of value are included in the model.

2.4.2 _ Defining Subjective Quality Variables

In addition to the list of property characteristics, implementers are very likely to want to include variables for the quality of different characteristics. For example, there can be large differences, in terms of property value between similar buildings with “good” or “poor” quality zinc roofs, and capturing those differences is often essential for the accuracy of the final model.

The challenge with including quality variables in the model is that they are inherently subjective. Different enumerators might reasonably have different interpretations of a “good quality” or “poor quality” zinc roof. This entails the need for clear definitions—and subsequent training—that allow enumerators to assess quality within building characteristic categories. For instance, when considering roof quality, enumerators must be able to distinguish a “poor” or “good” quality zinc roof *in comparison to all other zinc roofs in the jurisdiction* (rather than in comparison to other building materials, like concrete or tile).

As much as possible, quality definitions should emphasize the observable aspects of property characteristics. For instance, a “poor quality” zinc roof might be defined as one with visible holes, or when tires, rocks or other material are used to secure the roof in place, while a “good quality” zinc roof might be defined as one that appears relatively new with little or no visible rust.

2.4.3 _ Iteration and Field Testing

Once the initial list of property characteristics is created, field testing and iteration are required to reach consensus on the specific definitions of property characteristics, and to ensure that they can describe all properties in the jurisdiction with sufficient accuracy.

To field test the effectiveness of the property characteristics list, a short interim survey can be deployed, with enumerators independently collecting data on the same properties. Analysts can then examine the data to identify characteristics with a high degree of discrepancy between enumerators, where further discussion and clearer definitions are required. This process of iteration and field testing should be repeated until a pre-defined level of consistency is reached between the enumerators. As described in Section 3.5, data should be similarly analyzed throughout the survey process to pro-actively identify problematic characteristics where more discussion or training are needed.

2.5 Additional Data

Aside from collecting information on components of the valuation model, most jurisdictions will want to collect additional data about residents, owners, and businesses as part of the property survey. For instance, governments may want to know the tenure status of properties, ownership, the contact details of residents and owners, or the types and registration status of businesses in the jurisdiction. While these data are unnecessary for implementing the simplified valuation model, they can have a wide range of uses for different levels of government.

In general, knowing the name and contact details of residents or owners will be useful when communicating with taxpayers and establishing trust in the credibility of the tax system. At a minimum, in most cases implementers will want to collect data on the names of residents, their relation to the property (e.g., owner, tenant, or other), their contact details, and the colloquial address of the property.

The challenge for implementers of a simplified valuation system is that collecting this type of additional data requires face-to-face interaction between enumerators and residents, and the cooperation of respondents. While collecting data on property characteristics for the points-based valuation can be accomplished through simple external observation, recording the name and contact details of property owners requires enumerators to speak with someone at each property and convince them to provide the needed information. This can slow the data collection process, especially due to frequently absent residents or those with limited trust in government.

Implementers will need to carefully consider the trade-offs in collecting these types of additional data and, where necessary, strike a compromise between comprehensiveness, accuracy, and expediency.

For instance, in many jurisdictions, identifying property ownership is a complex and arduous task. Residents may be reluctant to disclose property ownership and, in some cases, ownership may be contested. A compromise solution would be to instruct enumerators to collect whatever contact details are immediately available when visiting a property without requiring validation. The onus can then be placed on property owners to correct any errors in contact details once the reformed property tax system is implemented and tax bills have been issued, or the government could set up follow-up processes through which to progressively improve the completeness and accuracy of this information.

3 Implementing the Property Survey

Once the inputs to the valuation model have been clearly defined and validated through iterative field testing, the next step is implementing the property survey. The goal of the property survey is to visit every property in the jurisdiction and collect all information necessary to assess them using the simplified model. The property survey may simultaneously aim to validate information in the geospatial register⁹ and in the GIS layers created to define locations, special zones, or neighbourhood features, or to collect additional data of value to the process but beyond the scope of the valuation model itself (see Section 2.5).

3.1 Preparing Data Collection Maps

To organize data collection, the jurisdiction needs to be subdivided into manageable enumeration areas (to be assigned to teams) and smaller enumeration zones (to be assigned to individual enumerators).

⁹ *Strategies for validating the geospatial register are discussed in the Guidance Note: Property Identification and Mapping for Property Taxation.*

Each enumeration area and zone should cover roughly similar numbers of properties. If the geospatial register is complete, it provides a count of properties that can be used to organize roughly equal enumeration areas and zones. If the geospatial register is incomplete, open-source data¹⁰ can be used to estimate the number of properties in the jurisdiction and subdivide enumeration areas accordingly. Subdividing this way helps structure and monitor data collection so that no properties are missed and data collection proceeds efficiently.

The size and number of enumeration areas and zones will depend on the number of field personnel available for the exercise and the intended timeframe of data collection. Prior experience suggests that, as a conservative estimate, enumerators working in a city centre can expect to complete about 20 property surveys per working day. This rough average pace, coupled with the number of properties identified in the jurisdiction (either from the geospatial register or from open-source data), can allow implementers to estimate the number of enumerators required for the survey exercise and the expected timeframe for completion.

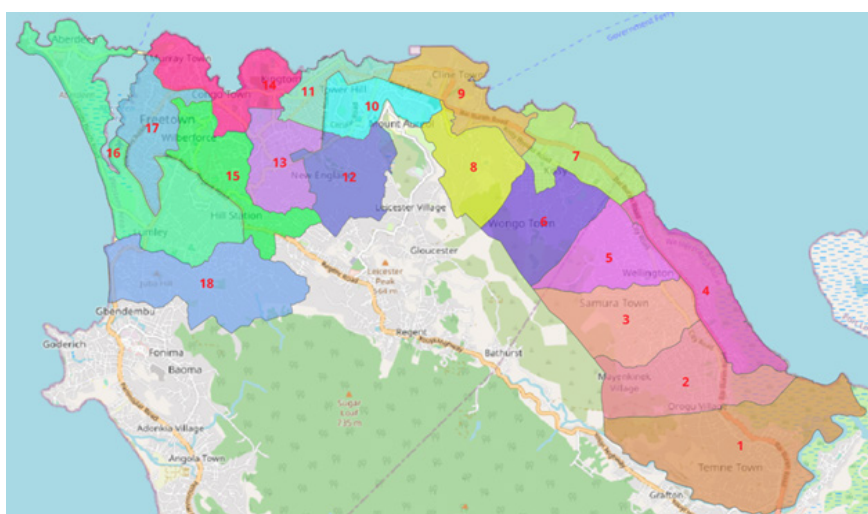


Figure 3 Enumeration areas for 18 data collection teams in Freetown, Sierra Leone

When creating enumeration maps, it is worth checking whether maps of administrative boundaries or other geospatial divisions already exist. Potential sources are the United Nations Office for the Coordination of Humanitarian Affairs (UN-OCHA) website,¹¹ or governmental institutions such as national statistics or geography departments. Alternatively, manual divisions can be created in the GIS software following either known neighbourhood borders or natural boundaries and major streets. These pre-defined areas can be used as a base to create enumeration areas that are relatively intuitive and easy to navigate for field staff. Enumeration areas should be geographically coherent so that teams do not need to travel to different areas during survey implementation. Enumeration areas can be further subdivided into smaller enumeration zones (i.e., for work to be completed by a single enumerator) to help ensure more rigorous monitoring of data quality, and that no properties are missed.

¹⁰ Open Street Maps, Google Open Buildings, and Bing Building Footprints are all viable data sources for this purpose. Although building footprint data from open-source platforms are often incomplete and somewhat inaccurate, they are usually sufficient to calculate a rough estimate of the number of properties in a jurisdiction.

¹¹ <https://data.humdata.org/dataset>



Figure 4 Enumeration zones for enumerators on team 11 in Freetown, Sierra Leone

Once enumeration areas and zones are created, they need to be merged with the geospatial register. Through this merge, enumeration areas and zones are matched with property GPS coordinates so that teams and enumerators can navigate to the properties allocated to them for data collection.



Figure 5 Rooftop polygons merged with building GPS coordinates in Beira, Mozambique

3.2 Survey Team Composition and Management

How data collection teams are organized is flexible, especially as available personnel and skillsets, budgets, and local circumstances will vary between contexts. This guidance note does not aim to prescribe an organizational structure for data collection. At a broad level, it is useful to describe the key tasks and responsibilities that need to be fulfilled for a successful and efficient data collection process.

Enumerators form the basis of the data collection team, as they are the ones actually gathering information on every property. Their main responsibility is to collect high-quality and accurate data for every property using the survey instrument.

Enumerators are grouped into teams each led by a supervisor. Supervisors are responsible for organizing their group of enumerators and ensuring their enumerators' work meets defined standards. This role can be systematized by having supervisors administer a spot check survey that assesses the ability of their enumerators to follow protocols (described in Section 3.5). Supervisors are also responsible for reporting on progress and any challenges encountered to the field coordinator.

At the top of the field teams sits one (or several) field coordinator(s). On one hand, the field coordinator is the extended arm of the project team and communicates any information or feedback to the team supervisors. On the other hand, the field coordinator gathers and reports information from the field back to the project team so that they are aware of what may not be visible in the data (e.g., complications with some property characteristics, reluctance from citizens to respond, etc.).

The project team should include at least one data analyst responsible for monitoring data quality in real time, including through the implementation of high-frequency checks (described in Section 3.5). The project team is responsible for quickly raising any potential data quality issues with the field coordinator(s) so that they can be addressed early in the data collection process. The field coordinator(s) are responsible for ensuring that any changes to the survey required to maintain data quality are implemented appropriately by the supervisors and enumerators.

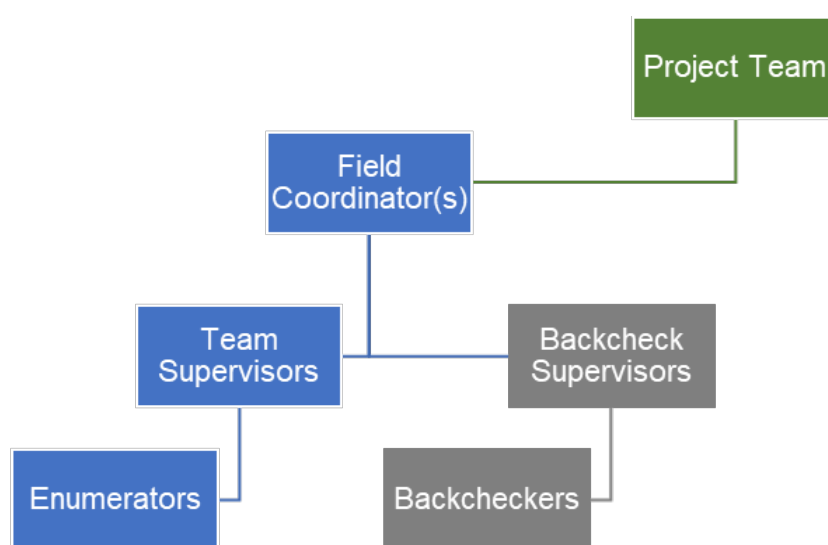


Figure 6 Potential organizational chart for data collection roles

Aside from the core field team (the blue roles in Figure 6), the survey should include a parallel structure of roles to monitor data quality (grey roles in Figure 6). These roles are intended to monitor the data collected by enumerators in real time so that any potential issues can be addressed quickly. While including these quality control roles in the data collection increases costs in the short term, they also help mitigate the risk of major data quality issues that can be very expensive to address once the survey is complete (e.g., if an error in data collection requires revisiting every property).

The foundation of the quality control team are backcheckers who are tasked with collecting the same data as enumerators from a proportion of properties that have already been visited.

Analysts can then compare data collected by backcheckers with data collected by enumerators to identify potential discrepancies. Using backcheck data for quality control purposes is described in more detail in Section 3.5 below. Like enumerators, teams of backcheckers answer to a backcheck supervisor who is responsible for communicating both ways between their backcheckers, the field coordinator(s), and the project team, and for ensuring the performance of their backcheckers.

3.3 Mobile Application Uses and Considerations

To administer the property survey, the use of a data collection mobile application is recommended. Depending on the IT system used to administer property taxes, a native mobile application may be available for data collection. Otherwise, there are several third-party applications on the market that can be used to develop simple data collection tools.¹² One advantage of a native mobile application is that it may be fully integrated with the existing IT system and allow for data to be formatted appropriately and synchronized automatically with the core property tax system.¹³ Using third-party applications may require more extensive data cleaning, restructuring, and a clear plan to migrate data into the core IT system once the property survey is complete. Ultimately, the choice between a native or third-party data collection app will be guided by the specific costs and benefits in specific locations and the choice of the IT system to administer property taxes.

Whatever solution is preferred, there are several basic requirements that a mobile data collection tool should meet to be effective:¹⁴

- The survey tool needs to offer unique logins for different team members, so data can be allocated to individuals and traced back to the same individuals for quality control.
- The survey tool—in combination with the device used—must enable photos and GPS coordinates to be recorded and stored in the application. GPS is required for enumerators to navigate to their allocated properties, and to confirm that they visited the correct properties. The value of property photos is explained in Section 3.4 below.
- To navigate the enumeration area, the survey tool also needs to accommodate either basic maps or aerial imagery. These maps provide a visual representation of where allocated properties can be found and can serve to validate the rooftop polygons drawn for the geospatial register.¹⁵
- Ideally, the survey tool can also be used entirely offline. Offline use permits teams to collect data in areas where there is poor internet connectivity and decreases the costs of the survey because mobile data is not required for enumerators throughout the day.

¹² Popular data collection mobile application providers include the open-source ODK and KoBo Collect, and the commercial SurveyCTO. SurveyCTO offers higher levels of data security but comes at a significantly higher cost for large surveys.

¹³ Moreover, a native mobile data collection app allows for the continuous discovery and inclusion of new or altered properties after implementation of the reformed valuation system.

¹⁴ A more detailed description of these requirements can be found in the Guidance Note: Key Elements of IT Systems Design for Property Taxation.

¹⁵ For more details on validating the geospatial register see Guidance Note: Property Identification and Mapping for Property Taxation.

3.4 Importance of Collecting Property Photos

Taking property photos during data collection should be considered an essential part of the survey in most cases, as they provide substantial future value for navigation, building trust in the simplified valuation system, and for adjudicating appeals to the assessed value of properties.¹⁶

When paired with rooftop GPS coordinates, photos help ensure that the same property can be located again. Locating specific properties is essential for backcheckers to ensure the quality of data collected, and for the subsequent delivery of tax bills or any additional visits to the property (e.g. for enforcement efforts). Photos are especially helpful for locating specific properties in areas without clear or consistent street addressing systems.

In addition to navigation, collecting property photos is useful to help build taxpayer trust in the simplified valuation approach. For instance, including property photos on tax bills helps to reassure taxpayers that they have been issued a bill for the correct property.

Property photos also have substantial value for both administrators and taxpayers during the appeals process. If taxpayers believe that an error has been made in their assessment, then property photos may help to quickly adjudicate their appeal. For instance, if an enumerator recorded that a property has concrete walls, but the owner argues that their walls are in fact lower-value brick, the validity of the appeal can be adjudicated with the property photo. This will help avoid time-consuming and potentially expensive field visits.

For some residents, taking property photos can be a sensitive issue. It is essential that enumerators are trained to reassure residents that property photos will be kept secure and will only be used for navigation and data verification purposes. Furthermore, extensive sensitization campaigns facilitate the enumerators' ability to collect property data, including photos.

3.5 Strategies to Ensure Data Quality

Ensuring that the simplified valuation system produces fair, transparent, and consistent property assessments rests on the quality of data collected during the property survey.

Implementers have several strategies at their disposal to help make sure that data collected are of the highest quality. To avoid expensive and time-consuming delays to clean data or revisit properties, it is important that these strategies are built into the data collection process from the beginning.

Ensuring that high-quality data is collected begins from training enumerators, supervisors, backcheckers, and field coordinators on the same protocol. Even staff with substantial expertise in valuation, construction or data collection should complete the training to ensure that everyone has a consistent understanding of the survey. Field staff can also be involved in the iterative testing of the property characteristics list and the survey instrument, as their participation can often help to identify problems in the definitions of building characteristics

¹⁶ A technical challenge that implementers may face when collecting photos is the relatively large file size of pictures taken by most handheld devices. Large file sizes can be addressed either by adjusting the settings on the handheld devices used for data collection, when downloading the data for analysis, or when uploading data to the IT system. In most cases, a relatively small file size (~100 kB) will be sufficient for property identification, navigation, and potentially displaying on tax bills.

or the survey logic. It is simpler and cheaper to address these problems in the training stage, instead of after the survey has been deployed.

Once data collection begins, clear lines of communication ensure that problems encountered in the field are quickly communicated up the chain to the project team, and that problems revealed in initial data analyses are communicated back to enumerators in the field. A clearly articulated organizational hierarchy with regular check-in meetings throughout the survey exercise helps to ensure this two-way flow of information.

Team supervisors should be responsible for conducting spot checks on their enumerators. Spot checks are unanticipated visits where supervisors check that their enumerators are surveying the correct properties, and collecting data and answering questions correctly. Spot checks can be conducted informally, or with a separate spot check survey that asks the supervisor to answer questions about the enumerator's performance. Results should be communicated on a frequent (daily or weekly) basis to the field coordinator, and subsequently to the project team.

The survey can also be programmed to automatically collect background data. This background data can be used to monitor survey performance and quickly detect any anomalies or inconsistencies. For instance, automatically recording the start and end time of every survey helps to monitor the working hours of enumerators, while also providing a proxy for whether the information was collected truthfully. If the duration of an enumerator's surveys is significantly lower than the average, this might indicate that the enumerator is not conducting proper assessments of their properties. Similarly, collecting background GPS coordinates when an enumerator starts or ends a survey, and when they take a picture of the property, helps to determine if the enumerator was close enough to the intended building. Although caution must be exercised when interpreting background GPS coordinates (they are not always perfectly accurate), they can be valuable indicators of enumerators' behaviour and can help identify remote assessments.¹⁷

Data collected by backcheckers is also a crucial component of the data quality monitoring and assurance strategy. Backcheckers collect the same information as enumerators from a randomized sub-section of properties that have already been visited. Backcheck data is then compared to the main survey data to identify teams or enumerators that may require retraining.

To help ensure data quality, about 10% of properties should receive a backcheck. This proportion can be increased at the beginning of data collection, when enumerators are still in the learning phase and may be forming bad habits or assimilating incorrect property characteristic definitions. The proportion can again be increased near the end of data collection, when enumerators may be rushing their work as they see the end of fieldwork approaching.

Early in data collection, backcheck properties can be selected completely at random. Once enough surveys and backchecks have been completed for initial analysis, backchecking should oversample from the poorest performing enumerators and teams to ensure that they are receiving a more rigorous quality control check to detect any potential problems. The project team should therefore periodically (e.g., daily or every few days) assign backcheckers a new set of properties to visit.

¹⁷ Remote assessment in this context could mean that an enumerator remains in a constant place while "collecting" data on multiple properties that are far away. In this case, the background GPS location remains approximately constant while the location of the properties assessed are distant from the background location.

It is also important to keep backchecks blind by not providing backcheckers with the data that enumerators have already collected on the same properties. Keeping backchecks blind helps avoid potential problems with confirmation bias or collusion between enumerators and backcheckers that would undermine the quality control process.

Once backcheck data has been collected, analysts must decide how it will be compared to the data collected by enumerators. That is, if discrepancies are detected between enumerator and backchecker data, who is considered correct? The simplest approach is to consider backcheck data correct in all circumstances, unless clear evidence emerges to the contrary (e.g., from property photos). The downside is that there is no guarantee backcheckers have collected the correct data. If this strategy is pursued, then backcheckers need to be more highly trained and familiar with the survey instrument than enumerators. To achieve this, during training backcheckers can be selected from the highest-performing and most motivated enumerators.

Alternatively, a more rigorous—but also more resource intensive—approach is to send a third super-enumerator or super-backchecker to collect data from properties where there is an identified discrepancy. This third survey can be used to corroborate the data collected by either the primary enumerator or backchecker, and to decide which data will be used in the final valuation model. If specific teams or enumerators have consistently high rates of backcheck discrepancies, it may be necessary to re-do data collection for their areas and potentially dismiss the poorly performing enumerator(s) or teams.

In addition to backchecks, high-frequency checks (HFCs) should be core to any data quality strategy. HFCs are analyses run on incoming data at regular intervals (ideally daily) to test for irregularities. Regular HFCs serve several purposes, such as helping to identify and correct errors in data collection, to monitor survey progress and enumerator performance, and to detect any cases of data fraud. HFCs can be used to test if some survey questions are skipped more than others, or if some questions have many more outlier responses than others. Positive results might indicate that there is an error in the survey coding that needs to be corrected. HFCs can also be used to check the distribution of different responses between enumerators and teams. For instance, if one enumerator or team indicates that most roads are “high quality,” while all other enumerators or teams indicate that most roads are “average” or “low quality”, it may indicate the need for additional training. HFC results are not definitive, but provide evidence for parts of the survey that may require additional discussion or clarification during the data collection process. Ideally, HFCs should be conducted by a data analyst on the project team, with results and any issues arising communicated to the field coordinator(s) for action.

Despite all these quality control strategies, as with any survey, it is likely that some variables will have higher backcheck discrepancy rates than others. This is especially the case for more subjective variables like construction quality. Comprehensive training and quality control can minimize but rarely eliminate the discrepancy rate in these subjective variables. As described in Section 5.2, it is possible to address high discrepancy rates when preparing the data for analysis—but it is always better to identify and correct problems with the variables early in the piloting, training, or data collection stages of the project.

4 Implementing an Expert Value Survey

Collecting data from the property survey, coupled with roofline measurements from the geospatial register,¹⁸ provide the inputs necessary to generate property value estimates with a simplified points-based system. The next step is to develop the model itself, which determines how much any given property characteristic or feature is associated with an increase or decrease in value, and thus translates property characteristics into taxable values.

This process could, in principle, be done manually by a group of experts who, based on their professional judgements, assign “points” to different property characteristics (e.g., all else equal, properties with concrete walls receive 75% more points compared to tin walls). However, such processes are prone to inaccuracy and contentiousness and create scope for collusion and rent-seeking.

It is preferable, instead, to rely on a data-driven process to develop the valuation model based on more objective criteria. The first step in calibrating the valuation model is to collect data on market value estimates for a representative sample of properties. These market value estimates, together with data on the sample’s property characteristics and features, are used to calibrate the valuation model in a way that aligns the points assigned to different characteristics with actual values in the market.

Implementers can adopt a range of methods to collect market data on a sample of properties. Conceptually, the true market value of a property can only be known when it changes hands on the market at a specific price. In theory, the most accurate method to calibrate the valuation model would be to rely on a sample of properties that have recently been bought or sold, and for which the market price is known. In practice, however, this level of detailed real estate market data is rarely available. And even if some level of real estate market data is available, it is unlikely to be (1) recent enough to capture current market values, or (2) representative of all property types and value distributions in the jurisdiction. In most cases, instead of relying on real estate market data, the most practical approach to securing recent value estimates for a representative sample of properties is to deploy an expert property value survey. An expert property value survey asks a team of experienced property valuers to estimate market values for a randomly selected sample of properties.

4.1 Selecting Properties for the Expert Value Survey

The properties that experts assess need to be randomly selected from the population of all properties in the jurisdiction. Random selection helps ensure representativeness, such that the small sample of properties used for calibrating the model can act as a proxy for all properties in the jurisdiction.

The first step in implementing an expert value survey is deciding how many properties need to be sampled, and the sampling strategy that will be used. Survey sample sizes are typically calculated with a standard formula that considers the size of the entire population, the target margin of error (i.e., the accuracy of results), and the target confidence level (i.e., the probability that results are correct). The challenge with using this standard formula is that the sample selected for the expert value survey should be representative of every property

¹⁸ Guidance Note: Property Identification and Mapping for Property Taxation.

characteristic in the jurisdiction. That is, the sample needs to be representative of all wall, roof, and window types, and every other property characteristic used in the analysis.

Simulations run on property data from Freetown, Sierra Leone, and Kampala, Uganda, suggest that an ideal sample size would be around 5–10% of all properties to capture sufficient variation in all property characteristics. However, achieving this ideal sample size will often be unrealistic or impractical given its high costs. To date, applications of the points-based valuation system have achieved reasonably accurate results with smaller sample sizes of roughly 2% of all properties in the jurisdiction. In general, the sample size required for accurate results also depends on the relative homogeneity of properties in a jurisdiction. In jurisdictions where most properties are very similar, a smaller sample size is less likely to yield inaccurate results. In contrast, jurisdictions with a lot of variability in property types and construction styles will require a larger sample size to maintain accuracy.

The trade-off with using smaller sample sizes is that some rare property characteristics—which can be important drivers of value (e.g., swimming pools)—may be poorly represented in the data. Analysis of their effect on property value can then become unreliable. To counteract this trade-off, governments can intentionally oversample properties with key rare characteristics for the expert value survey. This approach relies on pre-existing knowledge or data to help identify properties with the key rare characteristics for inclusion. Alternatively, Section 5.2 discusses several strategies to deal with rare property characteristics when preparing data for analysis.

Ideally, the sample for the expert value survey should be drawn after the geospatial register and property survey are complete. If all properties in the population have been identified (in the geospatial register) and their characteristics and features enumerated (in the property survey), it is much easier to draw a random sample that is representative of the whole population of properties. The list of properties can be sorted by location and major property characteristic, and a proportional number of properties can be randomly selected from each category in the list. For instance, if 20% of all properties have zinc walls, then the random sample can be selected so that 20% of properties in the sample have zinc walls. Similarly, if 10% of all properties are in a particular location, then the random sample can be selected so that 10% of properties in the sample are from that location. Properties that are randomly sampled from the list can then be identified in the field from their GPS coordinates, roofline polygons, and property photos. It is therefore desirable to complete the geospatial register and property survey before sampling properties for the expert value survey.

It may sometimes be necessary to carry out the expert value survey before the whole population of properties is known. For example, if the government wants to conduct a pilot exercise to develop a valuation model before scaling up to the whole jurisdiction. In such cases, open-source data can be used to roughly estimate the total number of properties in the jurisdiction to determine the needed sample size, followed by a “random walk” sampling strategy to select properties for the expert value survey.¹⁹

¹⁹ *Random walk protocols typically involve enumerators sampling properties in the field based on pre-defined rules, such as coin-flips or random-number tables. The downside of typical random walk protocols is that enumerators may be biased towards selecting properties that are easier to access. A more rigorous random walk variation generates a set of random GPS coordinates, subdivided within pre-defined data collection areas, and instructs enumerators to sample the property that is closest to each GPS coordinate.*

Whatever sampling strategy is employed, implementers need to consider trade-offs between representativeness and the resources available. In general, larger sample sizes are more likely to be representative, leading to a more accurate valuation model. But larger sample sizes also incur higher financial and time costs for the project. Implementers need to carefully consider the minimum sample size required to yield a valuation model of sufficient accuracy and consistency.

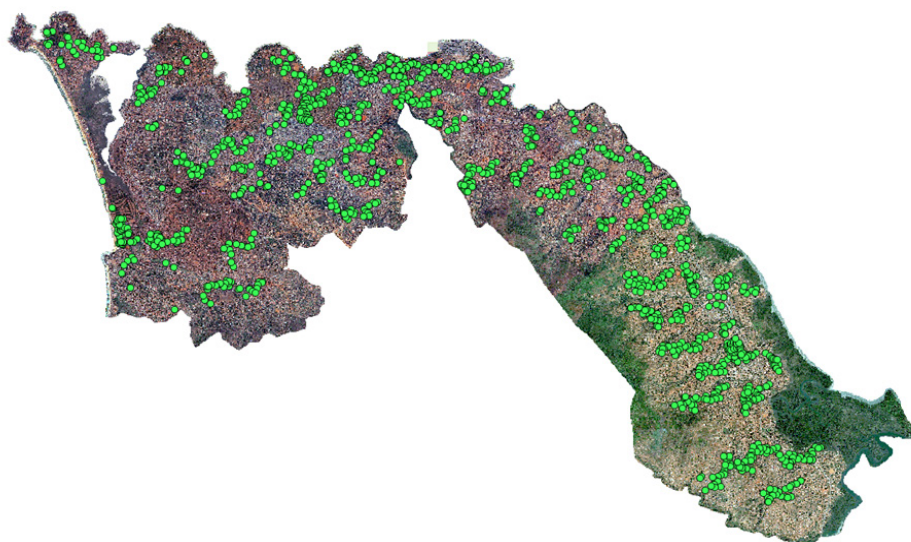


Figure 7 Sample of 2,000 randomly selected properties for the expert value survey in Freetown, Sierra Leone

4.2 Strategies to Collect Expert Value Estimates

Once properties have been sampled, expert valuers need to provide value estimates for each of the selected properties. This goal can be achieved using several possible methods.

First, implementers will need to decide which expert valuers to recruit for this exercise. Most countries will have a government department responsible for valuation, either at the central or local level. The expert valuation survey provides an ideal opportunity to involve these officials in the creation of the valuation model, and to position them to eventually assume responsibility for overall management of the simplified system.

Aside from government valuers, many countries also have an industry of professional property valuers working in private practice who can be recruited for the survey. Professional valuation expertise (both government and private practice) tends to be concentrated in major urban areas where property markets are most active. If the simplified system is being implemented in a smaller urban jurisdiction, professional valuers with knowledge of the local market may be very scarce. Implementers may therefore seek to work with real estate agents, chiefs, or other local officials with knowledge of the property market. When selecting experts for the property value survey, the main consideration should be sufficient knowledge of the local market to accurately estimate values for all property types in the jurisdiction.

If recruiting experts from different sources, they will likely employ different methods to estimate property values. Most professional valuers will use standardized formulas and approaches that may incorporate any market data available for the jurisdiction. Real estate agents or other local officials, by contrast, may have a more intuitive approach to estimating property values.

Combining these methods is not a problem for the simplified valuation model, provided the same approach is used to assess every property. It may be necessary to reach consensus among the valuers about which approach will be used, or alternatively (as described below) to average the estimates from different valuers to amalgamate methods and reduce variation.

Once experts have been recruited, the metric for estimating market value needs to be decided. Experts might be asked to estimate the price that a property would realize if sold on the open market (i.e., capital market value). Alternatively, rental value can be used as a proxy for market value. That is, experts can be asked to estimate the price that a property would rent for on either a monthly or annual basis.

Rental value is a proxy for market value to the extent that properties that rent for a higher value are also likely to sell on the open market for a higher value. Rental value may have the benefit of being easier to estimate in contexts where property markets are very opaque or inactive, and where most properties are tenanted. In many cases, the approach taken to estimate value will be dictated by local laws. For instance, legislation in some countries mandates the use of capital market value as the basis for property taxation, while other countries are more flexible and permit the use of proxies like rental value. Ultimately, the specific metric used is less important than ensuring that the approach used is consistent and comparable across all properties.

Implementers then must decide whether expert valuers will conduct their estimates through in-person property visits or through a desk-based exercise. For in-person assessment, experts are expected to visit every property in the sample to conduct their value calculations. For a desk-based exercise, valuers are provided with relevant information for each property such as location, size, characteristics, and features—along with property photos—to calculate their estimates. Desk-based assessment can be implemented as a simple survey to systematize the work and provide constant data feedback to the project team (an example is shown in Figure 8).

Desk-based assessment tends to be faster (and therefore cheaper), as experts are not required to visit each property in the field. In-person assessment is more resource intensive, as experts will require transport allowances and their pace will be slower. But in-person assessment may also be more accurate, especially for real estate agents or others who follow a more intuitive approach to estimating property values. In deciding on the approach to collecting property value estimates, implementers will need to consider the nature of the local property market, the level of capacity of the expert valuers, the methods they will employ to assess properties, and the availability of resources.

Several strategies can be deployed to cross-check the estimates produced by valuers, or to amalgamate their estimates if they are using slightly different methods. To allow for a degree of uncertainty in the estimates, experts can be asked to define a likely range of values rather than point estimates. For instance, valuers can be asked to define a lower and upper range at which a property is likely to sell (or rent) on the open market, as shown in Figure 8. These ranges can later be averaged to calculate a point estimate of property value. Multiple different valuers can also be asked to revisit the same properties and conduct separate assessments. Ideally, each property in the sample would be assessed by a minimum of two valuers to provide a second opinion—although doing so is dependent on available resources.

Rental Values

Go to

Floors: 0.25 floor(s)
Surface Area: 355.5 sq feet
Street Paved: No, in Poor condition, No street lighting
Street Category: Minor with One Lane, with the property having a(n) Easy access to the street
Drainage: Open
Features: No features
Potential to Build: Medium
Customer Parking: 2-5
Public Parking: None
Water: No Water
Grade of Design: Traditional
Ground Floor Use: Domestic
Wall: Masonry with Painted or Whitewashed and in Average condition
Veranda: Open
Fence: No fence
Security: No
Roof: Zinc / Metal Sheetting, in Average condition
Windows: Louvre
Aircondition: No
Garage: None
Outbuilding: No Outbuilding
Pool: No

Front

☐ Front


Side

☐ Side


Please enter your rental value estimates for the entire commercial property.

Lower Annual Rent Estimate

Your answer here...

Upper Annual Rent Estimate

Your answer here...

Currency

☐ SLL
☐ USD

Next >

Figure 8 Simple survey to collect expert value assessments in a desk-based exercise

To identify areas where a third or fourth opinion may be necessary, separate assessments of the same properties can be used to analyze the discrepancy rate between different valuers, and how discrepancies vary by property type and across the value distribution. Subsequent visits to the same property should be blind, such that valuers do not know which of their properties will be revisited, to reduce the risk of collusion between valuers. For the final analysis, estimates from different valuers can be averaged to reduce variation or, alternatively, if evidence suggests that some valuers are more reliable than others, their estimates can be preferred.

In addition to value estimates, inputs to the valuation model also need to be collected for each property in the sample (i.e., size, geographical variables, and characteristics). If the expert value survey is deployed after the geospatial register and property survey are complete, then these inputs are already known for the sample properties. If the expert value survey is deployed prior to or in parallel with the geospatial register and property survey, it will be necessary to collect all the same information from the sample properties. Data technicians can be employed to measure property rooflines from aerial imagery, and enumerators can collect data on geographical variables and property characteristics. The final result of the expert value survey should be a dataset of a representative sample of properties for which both the inputs to the valuation model and market value estimates are known.

5 Building the Valuation Model

Once compiled, a sample dataset of property information and estimated values can be used to build the valuation model. This involves generating a formula that translates the size, location, neighbourhood features, and characteristics of any property in the jurisdiction into a taxable value to which a tax rate can be applied.

The simplified valuation approach explained in this guidance note aims to create a model that is relatively accurate, simple to understand and easy to replicate. While the valuation approach overall strives to balance accuracy and simplicity, a basic fluency in statistics will help with understanding the mechanics underlying the model. When seeking to design and implement a simplified valuation model in specific contexts, experts or advisors with a relevant background in statistics will be required.

To explain the valuation model intuitively, the formula assigns “weights” to each property characteristic, with those “weights” indicating how much more or less valuable properties with that characteristic are, on average. So, for example, having high-value windows may have a “weight” of +35%, indicating that properties with high-value windows tend, on average, to be 35% more valuable than properties without high-value windows.²⁰ Technically, these weights are coefficients generated by the regression model. The terms “weights” and “coefficients” will be used interchangeably throughout this section of the guidance note.

5.1 Choosing the Statistical Model

Once the data is in hand, implementers need to decide on the econometric model that will be run to estimate the weight on each characteristic.

Relatively complex econometric or machine learning models can potentially be used to maximize accuracy. However, such complex models have significant downsides: they require high levels of statistical capacity to implement, and they can severely undermine comprehension and transparency for administrators and taxpayers.

The alternative is to use simpler models. An Ordinary Least Squares (OLS) model is among the simplest and most intuitive econometric models that can be used for this purpose. Existing evidence suggests that the accuracy gains from more complex models are relatively modest, and that simpler models like OLS are likely appropriate in most applied settings.

An OLS regression model estimates the coefficients of a linear equation that describes the relationship between several independent variables (i.e., the model inputs—size, geographical variables, and characteristics) and a dependent variable (i.e., estimated market value). At a conceptual level, an OLS model attempts to find the line of best fit—that is, the straight line that best describes the data. In this case, the data that the line of best fit attempts to describe are the property value estimates from the expert value survey. The coefficients for each property characteristic estimated by the model make up the linear equation that describes the line of best fit.

²⁰ Note that a coefficient of +35% does not mean that having high-quality windows *makes* a property 35% more valuable. Instead, having high-quality windows is a *consistent indicator* of higher-value properties, and so can be used as a proxy for estimating property value.

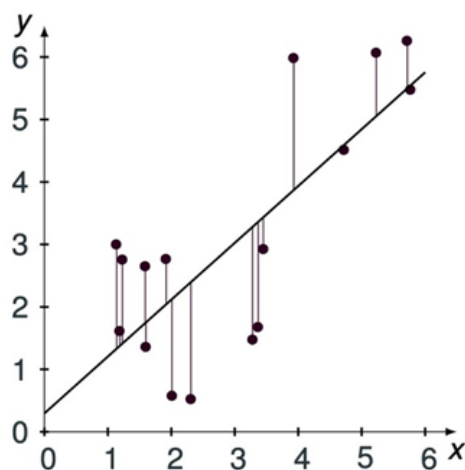


Figure 9 Line of best fit in an OLS model

An analyst can then look at the coefficients of the linear equation to determine how much—everything else held equal—a particular property characteristic is associated with an increase or decrease in estimated property value. In other words, the regression analysis says: after accounting for other factors, properties with X characteristic tend to be Y percent more or less valuable.

5.2 Preparing the Data for Analysis

Before running the OLS model to determine the coefficients, the data must be appropriately prepared, which can involve several trade-offs related to accuracy and statistical significance.

5.2.1 _ Dealing with rare property characteristics

Some characteristics collected during the property survey may be quite rare in the jurisdiction. A common challenge of regression-based analyses is that they struggle to yield useful coefficient estimates for rare characteristics because of small sample size.

If the geospatial register and property survey are completed before the expert survey, it is possible to select enough properties with rare characteristics for the expert value survey to ensure a sufficiently large sample for analysis (as described in Section 4.1). However, this will not be possible in all cases—either because the expert value survey needs to happen in parallel with the property survey, or because expanding the sample size to make sure enough of these rare characteristics are selected becomes prohibitively expensive.

An alternative approach is to deal with this problem in the analysis stage when preparing the data for the regression model. For example, some property characteristics—especially those with a high number of options in the survey—can be collapsed into a smaller number of options. Doing so reduces the number of coefficients that the model needs to estimate, increases the sample size per group, and simplifies the analysis.

For instance, the variable for “wall structure” may have initially been coded by construction type (e.g., brick, concrete, tin, wood, etc.). If some of these characteristic options are rare in the jurisdiction, too few may be selected during the expert value survey to yield useful coefficient estimates. To resolve this problem, the “wall structure” variable can be collapsed into fewer options defined by value (e.g., brick and concrete can be defined as “high-value material,” while tin and wood can be defined as “low-value material”). Collapsing characteristic options

in this way can help ensure a sufficiently large sample size in each of the categories for the regression analysis to yield reliable results.

It is still useful to define characteristics based on construction type during the survey, as construction type is easier to identify and objectively define for enumerators. But once the data is collected, variables can be recategorized into fewer buckets to ensure a larger sample size for analysis. Where none of these strategies is entirely effective, it may be necessary to also rely on expert opinions to adjust the weights given to certain variables, as described in Section 6.

5.2.2 _ Dealing with High Characteristic Noise

Some property characteristics may also have a high degree of noise in the data collected, reflecting the challenge of stipulating clear definitions for the subjective quality variables. That is, they might suffer from a high degree of variation unrelated to the actual characteristic but due to measurement error (e.g., different enumerators coding the same characteristic differently), causing the regression model to yield biased results. As discussed in Section 2.4, some of this noise can be minimized in the data collection stage with reference to concrete examples in the characteristic definitions and through extensive training, field testing, and iteration with the enumerators. But the inherent subjectivity of some characteristics means that a level of noise will likely remain in the final data.

Before running the regression model, the analyst should check the discrepancy rate of all characteristics by comparing them to backcheck data. While it is preferable to address high characteristic noise during data collection, it can also be dealt with during the analysis stage. One option is to dichotomize characteristics with a high degree of noise. For instance, a variable that was initially coded as having several options (e.g., none, small, medium, large) can be dichotomized to simply express whether the property characteristic is present or not (e.g., yes, no). Dichotomizing characteristic variables reduces the potential precision of the model, but coefficients become more reliable and the overall model simpler. Where characteristic noise is too high, certain variables might, alternatively, need to be excluded from the model as discussed in Section 6.

5.2.3 _ Dealing with Multicollinearity

Multicollinearity happens when two or more independent variables (i.e., property characteristics) that are highly correlated are included in the model. The model then has trouble distinguishing the impact of each individual variable on the estimated property value.

For example, if the property survey contains multiple variables related to road type (major, minor, footpath) and whether a road is paved, those variables may suffer from multicollinearity. In most cases, all major roads will be paved while footpaths will be exclusively unpaved. Including variables for both pavement status and road type may therefore be redundant if pavement status can be determined simply by referring to road type, and vice versa. It can still be useful to collect data on all these variables during the property and expert value surveys, as the correlations between these variables may not hold true in all contexts. However, once data have been collected, the analyst should run a cross-tabulation to determine the level of multicollinearity between variables.

To deal with the problem of multicollinearity, several variables can be amalgamated for inclusion in the valuation model. For instance, road status can be recategorized as the following:

- > “Good” if the road is paved, the condition is new, and it is a major road.
- > “Bad” if it is not paved, is in bad condition, and is either a minor road or footpath.
- > All remaining roads can be reclassified as “Average.”

Similar amalgamation can be done for other variables in the dataset that display high levels of multicollinearity. Amalgamating similar variables in this way simplifies the valuation model, while increasing the statistical significance of the generated coefficients. Definitions of amalgamated variables need to be expressed clearly so that future administrations can easily explain the valuation system to taxpayers, and easily collect data on any new properties that need to be added to the tax system.

5.2.4 _ Log Transformations for Continuous Variables

For OLS models, it is good practice to log transform continuous variables²¹ if their distribution is skewed.²² Log transforming these variables helps shift their distribution to align more closely with a normal distribution (a bell curve shape). Property market value estimates (whether capital or rental value) typically tend to skew heavily towards the left. That is, there is a much higher frequency of low-value properties than the average.

The left-skewed distribution of estimated rental values can be seen in the data from Freetown in the left panel of Figure 10. Log transforming rental value helps to partially correct this left skew. That is, instead of including rental value directly in the model, the analyst instead includes the logarithm of rental value. As seen in the right panel of Figure 10, the distribution of the log transformed rental value more closely matches a normal distribution. Property size is another continuous variable that tends to be left-skewed (i.e., there is a much higher frequency of small properties than the average) and should be log transformed prior to inclusion in the model.

Log transforming estimated market value means that the model used is technically called a “log-linear model”—an OLS model where the dependent variable (expert estimates of market value) is log transformed. Using a log-linear model means that the model will predict log transformed values for each property, rather than directly predicting market values. However, predicted market values can be computed by exponentiating the model’s output, as the exponential of a log transformed value is the value itself

²¹ A variable is said to be “continuous” if it can take an infinite number of values between a given interval. Property value, for instance, can take an infinite number of values between \$1 and \$1,000,000 and is therefore said to be continuous.

²² Technically, OLS models assume that the error term is normally distributed, and this is more likely to happen if the continuous variables in the model are also normally distributed.

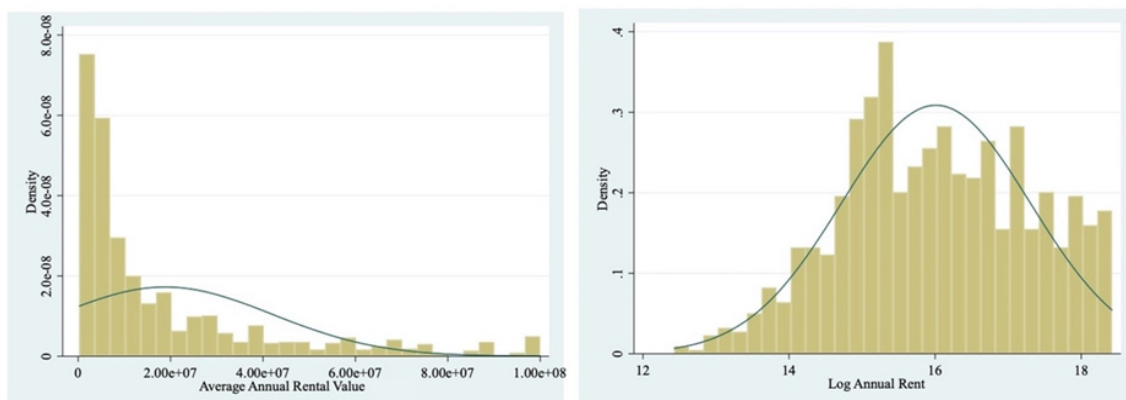


Figure 10 Log transformation of market value estimates

5.3 Interpreting the Valuation Model Output

Once the statistical model has been selected and the data prepared, a regression analysis is run on the data to generate a formula that can be used to produce property value estimates. In practice, the model works by first defining a “base property,” which is a property that has all the most common characteristics in the jurisdiction. Weights are then generated for each characteristic that describe how much that characteristic is associated with an increase or decrease in value relative to the base property.

	Characteristics	Weights	Significance
Street Quality	Bad	- 15 %	
	Average	+ 0 %	
	Good	+ 5 %	
Number of Lanes	None	- 20 %	
	One	+ 0 %	
	Two	+ 3 %	
	Four	+ 6 %	
Street Access	Easy	+ 0 %	
	Difficult	- 10 %	
Drainage	Yes	+ 2 %	
	No	+ 0 %	
Neighbourhood Features	Beach	+ 24 %	
	Environmental Hazard	- 15 %	
	Main Road with High Visibility	+ 18 %	
	Informal Settlement	- 21 %	
	Commercial Corridor	+ 25 %	
	None	+ 0 %	
Potential to Build	Yes	+ 5 %	
	No	+ 0 %	
Water	Yes	+ 5 %	
	No	+ 0 %	
Wall Material	Low-Value Material (Zinc)	- 50 %	
	Medium-Value Material (Mud, Wood)	- 32 %	
	High-Value Material (Masonry, Stone)	+ 0 %	

Wall Quality	Bad	- 10 %	
	Average	+ 0 %	
	Good	+ 25 %	
Roof Material	Low-Value Material (Asbestos, Aluminium, Zinc, Not Visible)	- 20 %	
	High-Value Material (Concrete, Tile)	+ 0 %	
Roof Condition	Bad	- 40 %	
	Average (and Not Visible)	+ 0 %	
	Good	+ 20 %	
Windows	Low-Value Material (Breeze Block, No Windows)	- 45 %	
	Medium-Value Material (Louvre, Traditional Glazed Casement)	+ 0 %	
	High-Value Material (Sliding Panels in Aluminium Frame)	+ 15 %	
Air Conditioning	Yes	+ 62 %	
	No	+ 0 %	
Security	Yes	+ 40 %	
	No	+ 0 %	
Outbuilding	Yes	+ 10 %	
	No	+ 0 %	
Pool	Yes	+ 150 %	
	No	+ 0 %	
Veranda	Yes	+ 8 %	
	No	+ 0 %	

	Confident: high level of statistical confidence and numbers broadly fit intuition
	Mixed: some statistical confidence but numbers broadly fit intuition
	Manually set weights

Figure 11 Characteristic weights for an example valuation model

Figure 11 shows an example of the valuation model output for a selection of property characteristics.²³ The model output has three components, each displayed in a column. The first column lists the property characteristics and features included in the model. Variables coded in grey are the base variables, meaning that they describe the most common property characteristic in the jurisdiction. The second column is the weight assigned to each variable that shows how much each property characteristic is associated with an increase or decrease in value. All the base variables (coded grey) have a weight of +0%, indicating that they describe the average property and thus receive no adjustment.²⁴ All other weights listed in the model are relative to the base variable of each characteristic.

²³ Note that, for simplicity, the model output in Figure 11 is a condensed version of a full model. A full model would likely include coefficients for building use, property type, and location.

²⁴ Technically, the base variables reflect the average value of the omitted category for each categorical variable.

The third column is the statistical significance of each weight, which describes how confident the analysis is about whether the weights are different from zero, and thus relevant for estimating property value. As described in Section 6.3, statistical significance is useful when making decisions about whether to keep or drop certain variables from the model. They are colour-coded, so that green represents a high level of statistical significance. Yellow represents some statistical significance but the coefficients broadly fit intuition. And light blue represents coefficients that were set manually, either because they are base variables or because their initial estimates had low statistical significance and did not fit intuition very well. Considerations when manually setting model coefficients are discussed in Section 6.4.

Walking through a specific example will help explain how the model output is applied in practice. In Figure 11, the variable “Roof Condition: Bad” receives a weight of -40%, meaning that if two properties are identical except one has a roof in bad condition and the other has a roof in average condition, their predicted rental values will differ by 40%. “Roof Condition: Average” receives a weight of +0% because it is the base variable against which all properties are assessed. “Roof Condition: Good,” in comparison, receives a weight of +20%, meaning that if a property’s roof condition changes from “Average” to “Good” (perhaps because it was recently replaced or repaired) the points assigned to that property by the model would increase by 20%.

6 Updating and Adjusting the Valuation Model

This valuation approach allows an analyst to generate a formula that takes property data as input, and outputs taxable values for all properties in the jurisdiction. The regression model provides a data-driven approach to answering the question: what is the relative taxable value of all properties in the jurisdiction? The regression-based approach to answering this question is attractive because it is data-driven and resistant to politicization, corruption, or simple poor judgement. However, there are several valid reasons why analysts or policymakers may wish to make manual adjustments to the valuation model.

Regression analysis using a limited sample is not always perfectly precise, and it may require manual adjustments if the model returns unintuitive results for weights assigned to property characteristics, or if the significance level of some weights is very low. Implementers may also manually include certain property characteristics or features in the model that are poorly captured by the regression analysis because they occur very infrequently. Policy priorities might also drive manual adjustments to the model—for example, to ensure that weights are transparent and easily understandable for taxpayers, to increase the overall equity of the model by overvaluing specific very high-value property characteristics, or to place greater importance on access to public services in shaping property tax assessments.

Once the regression analysis is complete, the model should enter a stage of policy review and discussion with local authorities and other stakeholders to incorporate manual updating and adjustment. This section describes some of the key compromises and policy decisions that can be made when implementing the model in practice, and their trade-offs. The simplified valuation approach is amenable to manual updating and adjustment because it provides a data-driven point of reference to understand the effects of any manual adjustments on overall model accuracy.

6.1 Measures of Fit as a Point of Reference

When looking at the regression output and considering any adjustments, a first point of reference is the R-square value, which provides a broad overview of the quality of fit of the model.²⁵ Technically, the R-square value indicates the percentage of variation in the dependent variable (market value estimates) that is explained by the independent variables (the property characteristics included in the model), and ranges from 0 to 1.

Measures of fit like the R-square value serve two purposes. First, they offer a general indication of the overall ability of the model to provide good estimates of property values. Most existing implementations of the simplified valuation approach have had an R-square value of 0.7 or above, offering overall confidence in the method. Finding an R-square value that was much lower—say 0.5 or below—might cause the analyst to revisit how the data was collected to explore possible problems that need to be addressed.

Second, the R-square value offers a guide or benchmark when analysts or policymakers wish to tailor the model by adding, dropping, or adjusting coefficients. At each stage of the discussion, the analyst can ask: “I might wish to make a particular choice to adjust the model, but let me first check how that would impact the overall quality of fit.”

Changes to the model that create a large decrease in fit should generally be avoided, while changes that have little impact on quality of fit but make the model more intuitive, clear, or able to meet policy goals can be safely adopted. The R-square value can therefore be used to ensure that decisions about any manual changes to the model are considered alongside their potential trade-offs on the model’s accuracy and reliability.

In addition to considering the R-square value, implementers can also analyze how manual adjustments to the model affect its relative ability to accurately predict property values across the value distribution. Some adjustments, for example, may tend to under-value higher value properties, while others may tend to over-value lower-value properties, with implications for the overall equity of the model. In general, manual adjustments that result in a significant decrease in overall model equity should be avoided.

6.2 Options for Calculating Surface Area Value

The most important driver of property value is the property’s size or surface area. Experience suggests that property value does not linearly increase as property size increases. That is, a property with 1,000 sq. ft. surface area is not, on average, twice the value of an otherwise identical 500 sq. ft. property. Instead, values tend to increase more slowly than surface area, on average. This relationship is shown graphically in Figure 12. The non-linear relationship between property size and value can be reflected in the model by raising surface area to the power of a coefficient. That coefficient determines how quickly, on average, value increases with surface area.

²⁵ While the R-square value provides a good first point of reference when considering model quality, there is also a risk of over-fitting the model to the calibration data. Over-fitting occurs when the model is tailored too closely to the details and noise of the calibration data, to the extent that it undermines the model’s performance on new, unseen data. As a result, while the model may perform well (i.e., have a high R-square value) on the calibration data, it will likely perform poorly when applied to the full property dataset. Strategies to mitigate overfitting include: a) simplifying the model as much as possible; b) regularization using Lasso or Ridge regressions that add a penalty for larger coefficients to discourage the model from fitting the noise; and c) cross-validation by splitting the sample into two (or more) datasets—the “training” data used to calibrate the model and the “new” data to test model performance.

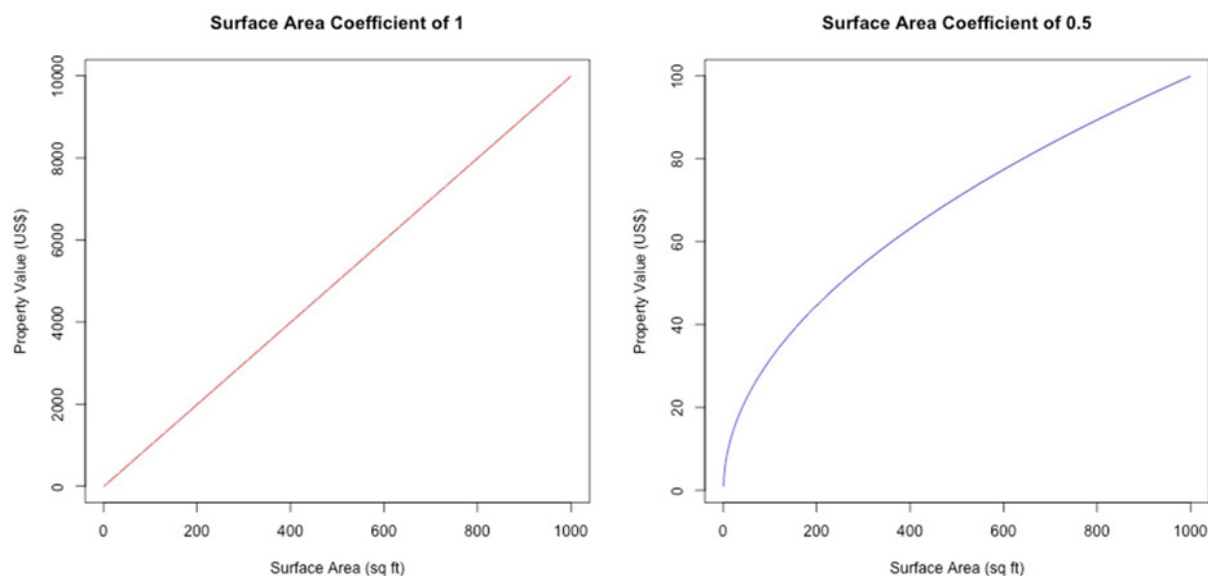


Figure 12 Relationship between property size and value for different surface area coefficients

In building a valuation model, implementers generally have three options for how to treat surface area:

1: Allow the surface area coefficient to be chosen by the regression model

Allowing the surface area coefficient to emerge from the regression analysis is the most data-driven and technically accurate approach. However, that coefficient is likely to be a very specific number—say, for example, 0.62—and thus unintuitive to interpret for taxpayers, and difficult to explain to those without a background in statistics. Alternatively, implementers can intentionally opt for a simpler coefficient that will slightly decrease overall model fit but may significantly increase public understanding of the valuation system. Here there are effectively two options:

2: Set the surface area coefficient equal to 1

The simplest and most intuitive approach is to manually set the surface area coefficient to 1, enforcing a linear relationship between property size and value on the model. While this approach is likely easier for taxpayers to understand, it will tend to overvalue properties with large surface areas (including, for example, large but unimproved older compounds) and undervalue smaller properties.

3: Set the surface area coefficient equal to 0.5

Raising any number to a coefficient of 0.5 implies taking the square root of that number. Square root is a more easily understood mathematical concept than more specific coefficients and is generally easier for taxpayers to calculate. However, using the square root of surface area will also reduce overall model fit, and may undervalue larger properties and overvalue smaller properties, depending on how far 0.5 differs from the specific coefficient assigned by the model.

When deciding how to treat surface area in the model, the analyst can be guided by how much the R-squared value changes depending on which coefficient is chosen. They can also look at how those changes affect the distribution of tax burdens across properties—for example, by making the model more or less progressive.

If setting the surface area coefficient equal to 1 causes only a small reduction in model fit, then it is likely a good policy choice given the benefits in terms of simplicity and transparency. If, by contrast, using either a coefficient of 1 or 0.5 causes a major decline in model fit, or seriously undermines the overall equity of the model, it may be appropriate to adopt the specific coefficient generated by the model that maximizes accuracy but sacrifices ease of understanding.

6.3 Dropping or Adjusting Model Variables

Before deploying the valuation model, an analyst may wish to drop or adjust specific variables if they do not contribute to the quality of the model, or if they yield unintuitive results. In deciding which, if any, variables to drop from the model, the analyst can be guided by the significance level of the variable coefficients and by the impact of dropping those variables on the quality of model fit.

In general, analysts may opt to drop variables in two circumstances, which may happen together:

- The variable lacks statistical significance, and dropping it has little impact on the quality of model fit. Any statistical analysis involves some level of uncertainty. Where a variable fails to achieve statistical significance or improve the fit of the model, the uncertainty and risk of corresponding administrative challenges of appeals is often high enough that it outweighs the benefits of retaining the variable.
- The statistical analysis returns a result that is unintuitive, and inconsistent with expert knowledge, and removing it has little impact on the quality of model fit. As above, any statistical analysis involves some level of uncertainty—and the possibility that a specific result is not an accurate reflection of reality. Such a result can be linked to problems with data collection, or simple chance, and is particularly true for variables with small sample sizes. Professional expertise and knowledge of the local property market can help identify such problems and make strategic adjustments to the model.

Instead of dropping problematic variables from the model, the analyst—in consultation with the specialized committee—may instead decide to manually adjust weights for variables that lack statistical significance or do not match intuition. The purpose of manual adjustment is to retain the variable in the model but ensure it matches a common-sense understanding of the drivers of property value. In making such adjustments the analyst can refer to their impact on the R-square value. In general, manual adjustments should be made large enough so that any unintuitive results are addressed (e.g., to reverse the coefficient sign) while maintaining the model's fit as much as possible.

Finally, implementers may wish to manually adjust certain coefficients to align with specific policy goals, or to increase comprehensibility for the public. To take a simple example, policymakers may worry about extremely high coefficients on specific variables leading to behavior change by taxpayers or eliciting backlash. Alternatively, policymakers may wish to make the overall valuation system more progressive by manually assigning larger weights to variables associated with very high-value properties (e.g., swimming pools), or alternatively by assigning negative weights to variables associated with very low-value properties (e.g., location in an informal settlement).

Care should be taken with such policy-driven adjustments, as they can be subject to political interference or can result in a loss of taxpayer trust if the overall valuation system is perceived as biased or unfair. As with other adjustments, it is important that decisions take account of the impact of such changes on overall model fit. Analysts may also ask whether such policy goals can be more effectively achieved using alternative tools. For example, if policymakers are seeking to increase progressivity, they could be better off adopting a progressive tax rate structure rather than adjusting the valuation model.

7 Calculating Taxable Values from the Model

Ultimately, the goal is to have a valuation formula that is both accurate and easily explained to both administrators and taxpayers (likely directly on tax bills). The formula has three key components: a base value, a surface area value, and all the value adjustments based on location, neighbourhood features, and property characteristics. The structure of that formula can be seen below:

$$\text{Taxable Value} = \text{Base Value} \times \text{Surface Area Value} \times \text{Total Adjustments Value}$$

The base value emerges directly from the regression analysis, is the same for every property, and can be interpreted as the estimated value of an average property in the jurisdiction with a size of one square foot.²⁶

The surface area value captures the fact that property value increases as property size increases. As discussed in Section 6.2, the surface area value may enter the model as a linear variable, or raised to an exponent that reflects the intuition that value often increases more slowly than property size, on average.

Finally, the different adjustments based on location, neighbourhood features, and property characteristics are multiplied together, as described below, to capture their overall upwards or downwards impact on taxable value.

Once the final valuation formula has been agreed upon—including any manual adjustments as described in Section 6—it can be used to calculate a taxable value for any property in the jurisdiction for which data has been collected. Figure 13 shows the application of the model to an example property, using information that might be communicated to a taxpayer on their bill. To calculate the taxable value assigned to a property (here expressed in points), the individual adjustments for each characteristic are multiplied together and applied to the product of a constant base value and the surface area value.

²⁶ Technically, the base value is the y-intercept of the line of best fit drawn by the OLS model.

Features	Result	Weight
Surface Area (in sq ft)	1200	34.64*
Building Type	Domestic	+ 0 %
Street Quality	Bad	- 15 %
Number of Lanes	One	+ 0 %
Street Access	Easy	+ 0 %
Drainage	No	+ 0 %
Neighbourhood Features	No	+ 0 %
Potential to Build	No	+ 0 %
Water	No	+ 0 %
Wall Material	Medium-value material	- 32 %
Wall Quality	Average	+ 0 %
Roof Material	Low-value material	- 20 %
Roof Condition	Average	+ 0 %
Windows	Medium-value material	+ 0 %
Air Condition	No	+ 0 %
Security	No	+ 0 %
Outbuilding	Yes	+ 10 %
Pool	No	+ 0 %
Veranda	Yes	+ 8 %
Location	Rating Area A	- 18 %
Constant (Base Value)	232400	232400



Base Value	232,400
Surface Area Value	34.64
Total Base Value	8,050,336**
Total Adjustment Value	-55%***
Assigned Points Value	3,622,651****

* = (Surface Area)0.5
 ** = Base Value x Surface Area Value
 *** = $\prod (1 + \text{coefficient}_n) - 1$
 **** = Base Value x Surface Area Value x Total Adjustments Value

Figure 13 Application of the simplified valuation approach to an example property²⁷

Calculating the taxable value of a property begins from the constant base value, which is the same for every property in the jurisdiction. In this example, the base value generated by the model is 232,400. The base value is multiplied by the surface area value of the property to determine the total base value of the property. In this example, the surface area value is set as the square root of the measured surface area of the property: $\sqrt{1,200} = 34.64$. Multiplying the base value and surface area value gives: $232,400 \times 34.64 = 8,050,336$. This total base value describes a property of 1,200 sq. ft. with all the most common property characteristics found in the jurisdiction.

But the property used in this example does not have all the most common characteristics, and thus needs to receive an adjustment. The property in Figure 13 receives adjustments for the following characteristics:

- Street Quality, Bad: -15%
- Wall Material, Medium-Value: -32%

²⁷ The figures in this model are similar to real-world applications but have been altered to maintain anonymity.

- Roof Material, Low-Value: -20%
- Outbuilding, Yes: +10%
- Veranda, Yes: +8%
- Location, Rating Area A: -18%

The remaining property characteristics listed in Figure 13 receive no adjustment because they are the base variables in this example model. To standardize the multiplication, each coefficient is first added to 1 before they are multiplied together:

$$(-0.15 + 1)(-0.32 + 1)(-0.20 + 1)(0.1 + 1)(0.08 + 1)(-0.18 + 1) = 0.45$$

This number means that the property displayed in Figure 13 is estimated to be 45% of the value of a base property of the same size. To express the total adjustment in percentage form, 1 can be subtracted from the product of the individual coefficients: $0.45 - 1 = -0.55$, or -55%. In other words, based on its characteristics and features, the property in Figure 13 receives a cumulative value adjustment of -55% compared to a base property of the same size.²⁸

The total adjustment value (-55%) is then applied to the total base value (8,050,336) to calculate the number of points assigned to the property: 3,622,651. A tax rate can then be applied to the assigned points to determine the bill that the property will receive.

If the information displayed in Figure 13 is clearly and transparently communicated on property tax bills, then in theory any taxpayer with a moderate understanding of mathematics and a calculator can compute these values themselves to understand how their tax liability was calculated. Such transparency can help inculcate a sense that the overall property tax system is fair, and that it is systematically applied to all properties in the same way. Enabling this high level of transparency is one of the strengths of the simplified valuation approach.

²⁸ The formula to calculate total adjustment value is displayed at the bottom of Figure 13 in the note marked: ***.

8 Appendices

→ Appendix 1 Approaches to Assessing Commercial Properties

The simplified valuation approach explained in this guidance note is applied easily to residential properties, which make up most properties in most jurisdictions. Commercial properties can present challenges to the simplicity of the model because their construction can be more idiosyncratic, making them more difficult to compare to each other and determine relative property values. In general, standard commercial buildings such as shops, workshops, or small warehouses can be easily incorporated into the simplified valuation model because their style of construction is roughly comparable to that of residential properties. For more distinct commercial properties, however, officials can choose between three broad strategies, each with its own trade-offs.

- **Adjust commercial property valuations within the core model based on relative construction costs.**

The simplest and most straightforward approach is to include commercial properties in the model as residential properties, with a subsequent adjustment based on their relative construction costs. For instance, the per-square-foot construction cost of a warehouse is lower than that of a residential property. Similarly, the per-square-foot construction cost of a bank is higher than that of a residential property. The model can therefore be calibrated to either discount or add to the estimated value of these commercial properties based on their relative construction costs. Relative construction costs for different building types can be extracted from international guides such as Marshall & Swift.²⁹ Despite its relative simplicity, this approach has several drawbacks. It uses the same basic valuation formula for residential and commercial properties, although (a) the factors driving the value of commercial properties may be different, and (b) a subset of commercial properties is very distinct in design and construction and may not be well represented by the standard survey (e.g., filling stations, hotels, industrial properties). Depending on the context and resources available, officials may wish to consider alternate strategies to assess distinct commercial properties.

- **Create a separate valuation formula tailored to distinct commercial properties.**

If the resources and capacity are available, then a separate survey and valuation formula could be developed that is specifically designed to take account of the unique features driving the value of commercial properties. This approach would ensure transparency and consistency across commercial valuations but would also be more subjective than the residential model because of small sample sizes (e.g., most jurisdictions have only a few hotels, and a few industrial facilities).

- **Continue to rely on expert valuation for distinct commercial properties.**

For a subset of unique commercial properties, it may make sense to exclude them from the model and continue relying on individual expert assessment. If most properties in a jurisdiction are assessed using a simplified AVM approach, then the number of unique commercial properties reserved for expert assessment will be small, and their revenue potential relatively high. If this approach is pursued, then it will be important to ensure that the methods used to assess unique commercial properties are made as transparent as possible to reduce subjectivity and increase taxpayer trust.

²⁹ Marshall & Swift is available commercially. Although construction standards may differ between markets, relative construction costs are likely to remain roughly consistent. See here: <https://www.corelogic.com/mortgage/appraiser-solutions/marshall-swift/>

→ Appendix 2 Generic List of Property Characteristics

Property Type
Commercial
Bank
Car Dealership
Petrol Station
Guest House
Hotel
Industrial Manufacturing
Industrial Warehouse
Mixed Retail / Office
Motor Garage
Office
Private Clinic
Private School
Retail
Super Market

Domestic
Institutional
Army Barracks and Installations
Court Buildings
Fire Station
Government Offices
Parliament Buildings
Police Buildings
Police Compounds
Municipal Offices

External Observable Property Characteristics
Street Paved
Yes
No
Street Condition
Like New
Average
Poor
Street Lighting
Yes
No
Street Category
Major
Minor
Footpath
Number of Lanes
No Lanes
One Lane
Two Lanes
Four Lanes

Street Access
Easy
Difficult
Drainage
Open
Covered
None
Potential to Build
None
Small
Medium
Large
Onsite Dedicated Customer Parking [in cars]
None
2-5
6-10
10+

Nearby Public Customer Parking [in cars]
None
2-5
6-10
10+
Water Supply
No Water
Municipal Water Supply
Tank
Well
Borehole
Grade of Design
Modern
Traditional
Use Ground Floor
Domestic
Commercial
Category of Construction - Wall
Masonry
Mud
Wood
Stone
Zinc
Wall Finish
None
Alucobond Metal
Glazed Curtain
Painted or Whitewashed
Stone
Tile
State of Repair
Good
Average
Bad
Veranda
None
Open
Enclosed

Fence and Gate Wall Construction
No Fence
Chain Link
Masonry
Mud
Palisade
Stone
Wood
Zinc
Fence and Gate Condition
No Fence
Good
Average
Bad
Security
Yes
No
Roof Type
Not Visible
Asbestos
Galvanised Aluminium
Concrete
Tile
Zinc / Metal Sheetting
Roof State of Repair
Not Visible
Good
Average
Bad
Windows (Dominant Types)
No Windows
Breeze Block
Louvre
Sliding panels with aluminium frame
Traditional glazed casement set in a metal frame
Wood
Air Conditioning
Yes
No

Garage
None
Inbuilt
Detached
Underground
Outbuilding
None
Generator
Outbuilding
Service Quarters
Storage Area
Pool
Yes
No

Locational Factors	
Administrative Zones	
Zone 1	
Zone 2	
Zone 3	
Commercial Corridor	
Yes	
No	
Informal Settlement	
Yes	
No	
Environmental Hazard	
Yes	
No	
High-value Area	
Yes	
No	

→ **Appendix 3 Template Identification Sheet for Externally Observable Property Characteristic**

→ Individual Property

Property N°: *number assessed*

Property ID: *ID assigned*

Insert picture of property here

Quintile (value) 1 to 5

Est. Rental
ValueLocation *ward*

Age *approx. Age*

Property Characteristics Description

[illegible]

→ Consolidation of All Property Characteristics

Assessable Properties

Property Characteristics

Category	Description
Wall Material	Mud
	Cement
	Zinc
	Stone
	etc.
Wall Quality	Good
	Average
	Bad
Locational Feature	Central Business District
	Beach
	Informal Settlement
	etc.
....	

Non-assessable

Description

Kiosk with non-permanent foundation
In non-construction area
etc.

